

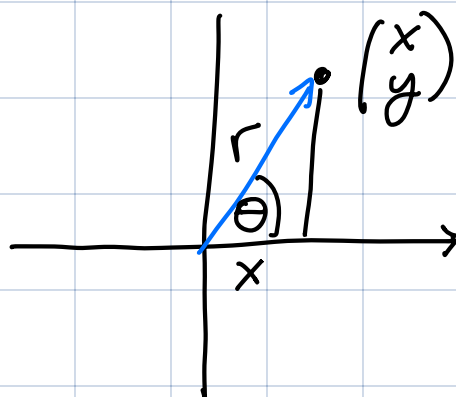
Lecture 12, 14.6.26

Remark 1: For every $\begin{pmatrix} x \\ y \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, there exist unique $r > 0$ and $0 \leq \theta < 2\pi$ s.t.

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$$

Note that $r = \sqrt{x^2 + y^2}$ and $\tan \theta = \frac{y}{x}$,
as long as $x > 0$ and $y \geq 0$.

Note also $\begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ iff $r \rightarrow 0^+$.



Exercise 1: consider the fun. defined by

$$F\begin{pmatrix} x \\ y \end{pmatrix} = \begin{cases} \frac{x^2 y}{x^2 + y^2} & \begin{pmatrix} x \\ y \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\ 0 & \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \end{cases}, \forall \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2$$

Determine whether F is cont. at $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$.
by using polar coordinates.

solution: we have

$$\lim_{\begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 0 \end{pmatrix}} F\begin{pmatrix} x \\ y \end{pmatrix} = \lim_{\begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 0 \end{pmatrix}} \frac{x^2 y}{x^2 + y^2} = \lim_{r \rightarrow 0^+} \frac{r^3 \cos^2 \theta \sin \theta}{r^2}$$

$$= \lim_{r \rightarrow 0^+} r \cos^2 \theta \sin \theta \stackrel{V \times B}{=} 0 = f\begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

thus F is cont. at $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$.



Example: consider the fun. defined by

$$F\begin{pmatrix} x \\ y \end{pmatrix} = \begin{cases} \frac{xy}{x^2+y^2} & \begin{pmatrix} x \\ y \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\ 0 & \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \end{cases}, \forall \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2$$

we have seen $\lim_{\begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 0 \end{pmatrix}} F\begin{pmatrix} x \\ y \end{pmatrix}$ does not exist

we check using polar coordinates:

$$\lim_{\begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 0 \end{pmatrix}} F\begin{pmatrix} x \\ y \end{pmatrix} = \lim_{\begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 0 \end{pmatrix}} \frac{xy}{x^2+y^2} = \lim_{r \rightarrow 0^+} \frac{r^2 \cos \theta \sin \theta}{r^2}$$

$$= \lim_{r \rightarrow 0^+} \cos \theta \sin \theta = \text{depends on } \theta.$$

Exercise 2: Let $a \in \mathbb{R}$ and consider the function defined by

$$F\begin{pmatrix} x \\ y \end{pmatrix} = \begin{cases} \frac{x^2 y + x y}{x^2 + y^2} & \begin{pmatrix} x \\ y \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\ a & \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \end{cases}, \quad \forall \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2$$

prove or disprove each of the following statements:

(a) there exists a value of $a \in \mathbb{R}$ s.t. F is cont. at $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$.

(b) there exists a value of $a \in \mathbb{R}$ s.t. $\frac{\partial F}{\partial x}, \frac{\partial F}{\partial y}$ exist at $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$.

Solution:

(a) we compute

$$\lim_{\substack{\begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\ y=0}} F\begin{pmatrix} x \\ y \end{pmatrix} = \lim_{x \rightarrow 0} \frac{0+0}{x^2} = 0$$

$$\sqrt{3+x^2} \quad \text{and} \quad \frac{x+1}{x^2+1} = 1$$

$$\lim_{\substack{(x,y) \rightarrow (0,0) \\ x=y}} F\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right) = \lim_{x \rightarrow 0} \frac{x}{2x^2} = \lim_{x \rightarrow 0} \frac{1}{2x}$$

thus, $\lim_{(x,y) \rightarrow (0,0)} F\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right)$ doesn't exist.

$\Rightarrow$ No value of a for which F is cont. at $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$.

draft: $\lim_{r \rightarrow 0^+} \frac{r^3 \cos^2 \theta \sin \theta + r^2 \cos \theta \sin \theta}{r^2} =$

$$\lim_{r \rightarrow 0} [\underbrace{r \cos^2 \theta \sin \theta}_{\downarrow 0} + \underbrace{\cos \theta \sin \theta}]$$

⑤

$$\frac{\partial F}{\partial x} \begin{pmatrix} 0 \\ 0 \end{pmatrix} \stackrel{\text{def.}}{=} \lim_{x \rightarrow 0} \frac{F(x, 0) - F(0, 0)}{x - 0}$$

$$= \lim_{x \rightarrow 0} \left[\frac{0 + 0 - a}{x^2} \right] = \begin{cases} 0 & a = 0 \\ \text{DNE} & a \neq 0 \end{cases}$$

$$a = 0$$

$$a \neq 0$$

thus $a=0$. Also

$$\begin{aligned}\frac{\partial F}{\partial y}\left(\begin{smallmatrix} 0 \\ 0 \end{smallmatrix}\right) &\stackrel{\text{def}}{=} \lim_{y \rightarrow 0} \frac{F(0, y) - F(0, 0)}{y - 0} \\ &= \lim_{y \rightarrow 0} \left[\frac{\frac{0+0}{y^2}}{y} \right] = 0\end{aligned}$$



Exercise 3: consider the fun defined by

$$F\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right) = \begin{cases} \frac{x^3 + y^2}{\sqrt{x^2 + y^2}} & \left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right) \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\ 0 & \left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right) = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \end{cases} \quad \forall \left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right) \in \mathbb{R}^2$$

(a) compute $\frac{\partial F}{\partial x}\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right)$ at every $\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right) \in \mathbb{R}^2$ or prove that it doesn't exist.

(b) compute $\frac{\partial F}{\partial y}\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right)$ at every $\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right) \in \mathbb{R}^2$ or prove that it doesn't exist.

Solution:

① First we compute $\frac{\partial F}{\partial x}\left(\begin{smallmatrix} 0 \\ 0 \end{smallmatrix}\right)$ by using the definition:

$$\frac{\partial F}{\partial x}\left(\begin{smallmatrix} 0 \\ 0 \end{smallmatrix}\right) \underset{\text{def}}{=} \lim_{h \rightarrow 0} \frac{F\left(\begin{smallmatrix} h \\ 0 \end{smallmatrix}\right) - F\left(\begin{smallmatrix} 0 \\ 0 \end{smallmatrix}\right)}{h} = \lim_{h \rightarrow 0} \frac{\frac{h^3+0}{\sqrt{h^3}} - 0}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h^3}{h \cdot |h|} = \lim_{h \rightarrow 0} h \cdot \frac{h}{|h|} \stackrel{\text{BxV}}{=} 0$$

For every $\begin{pmatrix} x \\ y \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, we have

$$\frac{\partial F}{\partial x}\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right) = \frac{3x^2 \sqrt{x^2+y^2} - (x^3+y^2) \cdot \frac{x}{\sqrt{x^2+y^2}}}{x^2+y^2}$$

thus

$$\frac{\partial F}{\partial x}\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right) = \begin{cases} \frac{3x^2 \sqrt{x^2+y^2} - (x^3+y^2) \cdot \frac{x}{\sqrt{x^2+y^2}}}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$$

⑥ First we compute $\frac{\partial F}{\partial y}\left(\begin{smallmatrix} 0 \\ 0 \end{smallmatrix}\right)$ by using the definition:

$$\frac{\partial F}{\partial y}\left(\begin{smallmatrix} 0 \\ 0 \end{smallmatrix}\right) \stackrel{\text{def}}{=} \lim_{h \rightarrow 0} \frac{F\left(\begin{smallmatrix} 0 \\ h \end{smallmatrix}\right) - F\left(\begin{smallmatrix} 0 \\ 0 \end{smallmatrix}\right)}{h} =$$

$$= \lim_{h \rightarrow 0} \frac{\frac{0+h^2}{\sqrt{h^2}} - 0}{h} = \lim_{h \rightarrow 0} \frac{h^2}{|h| \cdot h} = \lim_{h \rightarrow 0} \frac{h}{|h|}$$

$$= \begin{cases} 1 & h \rightarrow 0^+ \\ -1 & h \rightarrow 0^- \end{cases} = \text{doesn't exist!}$$

For every $\begin{pmatrix} x \\ y \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ we can use AOD

$$\frac{\partial F}{\partial y}\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right) = \frac{2y \sqrt{x^2+y^2} - 2y \frac{y}{\sqrt{x^2+y^2}}}{x^2+y^2}$$



Exercise 4: For each of the following fun., compute the first order partial derivatives at every point:

(a) $f(x,y) = x^2 y$

Solution: $f_x(x,y) = \frac{\partial f}{\partial x}(x,y) = 2xy$, $f_y(x,y) = x^2$.

(b) $f(x,y) = \sin x \cdot \cos y$

Solution: $f_x(x,y) = \frac{\partial f}{\partial x}(x,y) = \cos x \cos y$, $f_y(x,y) = -\sin x \sin y$

(c) $f(x,y) = x^2 y + xy^2 + \frac{1}{x} + \ln|1+x^2+y^2|$

Solution: $f_x = 2xy + y^2 - \frac{1}{x^2} + \frac{2x}{1+x^2+y^2}$

$$f_y = x^2 + 2xy + \frac{2y}{1+x^2+y^2}$$

(d) $\vec{F}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+y^2 \\ 2-xy-x \end{pmatrix}$

Solution:

$$\frac{\partial \vec{F}}{\partial x} = \begin{pmatrix} 1 \\ -y-1 \end{pmatrix}, \quad \frac{\partial \vec{F}}{\partial y} = \begin{pmatrix} 2y \\ -x \end{pmatrix}$$

(cost)

e) $f(t) = \begin{pmatrix} \sin t \\ \cos t \end{pmatrix}$

solution:

$$f'(t) = \begin{pmatrix} -\sin t \\ \cos t \end{pmatrix}$$

f) $f(x, y, z) = xy^2z^3$

solution:

$$f_x = y^2z^3$$

$$f_y = 2xy z^3$$

$$f_z = 3xy^2 z^2.$$



definition: the second-order partial der. of $\vec{F}$ is defined as

$$\frac{\partial^2 \vec{F}}{\partial x_j \partial x_i}(\vec{x}_0) = \frac{\partial}{\partial x_j} \left(\frac{\partial \vec{F}}{\partial x_i} \right)(\vec{x}_0)$$

we also denote $F''_{xy} = F_{xy} = (F_x)_y = \frac{\partial^2 F}{\partial y \partial x}$

Exercise 5: For each of the following fun., compute the second order partial derivatives at every point:

a) $f(x, y) = xe^y$

solution: we have

$$f_x = e^y, \quad f_y = x e^y$$

$$f_{xx} = 0$$

$$f_{yy} = x e^y$$

$$f_{xy} = \frac{\partial}{\partial x} (x e^y) = e^y$$

$$f_{yx} = \frac{\partial}{\partial y} (e^y) = e^y$$

$$\textcircled{b} \quad \vec{F}\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right) = \begin{pmatrix} x+y^2 \\ 2-xy-x \end{pmatrix}$$

solution:

$$\vec{F}_x = \begin{pmatrix} 1 \\ -y-1 \end{pmatrix} \quad \vec{F}_{xx} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \vec{F}_{yx} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$$

$$\vec{F}_y = \begin{pmatrix} 2y \\ -x \end{pmatrix} \quad \vec{F}_{yy} = \begin{pmatrix} 2 \\ 0 \end{pmatrix} \quad \vec{F}_{xy} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$$

$$\textcircled{c} \quad f(t) = \begin{pmatrix} \cos t \\ \sin t \end{pmatrix}$$

solution:

$$f'(t) = \begin{pmatrix} -\sin t \\ \cos t \end{pmatrix} \Rightarrow f''(t) = \begin{pmatrix} -\cos t \\ -\sin t \end{pmatrix}$$

Theorem 1. Suppose that $\frac{\partial^2 F}{\partial x_j \partial x_i}$ and $\frac{\partial^2 F}{\partial x_i \partial x_j}$ are
cont. at $\vec{x}_0$. Then $\frac{\partial^2 F}{\partial x_j \partial x_i}(\vec{x}_0) = \frac{\partial^2 F}{\partial x_i \partial x_j}(\vec{x}_0)$

proof: No proof. 😊

Exercise 6: consider the fun.

$$f(x,y) = \begin{cases} xy \operatorname{arctg}\left(\frac{x^2 - y^2}{x^2 + y^2}\right), & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$$

(a) compute f_x and f_y at every point.

(b) compute $f_{xy}(0,0)$ and $f_{yx}(0,0)$

(c) prove or disprove: Both f_x and f_y are
cont. at $(0,0)$.

Solution:

(a) we have for $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$, by definition

$$f_x(0,0) = \lim_{h \rightarrow 0} \frac{f(h,0) - f(0,0)}{h-0} = \lim_{h \rightarrow 0} \frac{0-0}{h} = 0$$

$$f_y(0,0) = \lim_{h \rightarrow 0} \frac{f(0,h) - f(0,0)}{h-0} = \lim_{h \rightarrow 0} \frac{0-0}{h} = 0$$

$\forall \begin{pmatrix} x \\ y \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, we use AOD,

$$f_x(x,y) = y \arctan\left(\frac{x^2-y^2}{x^2+y^2}\right) + \frac{xy}{1+\left(\frac{x^2-y^2}{x^2+y^2}\right)^2} \cdot \frac{2x(x^2+y^2) - 2x(x^2-y^2)}{(x^2+y^2)^2}$$

$$f_y(x,y) = x \arctan\left(\frac{x^2-y^2}{x^2+y^2}\right) + \frac{xy}{1+\left(\frac{x^2-y^2}{x^2+y^2}\right)^2} \cdot \frac{-2y(x^2+y^2) - 2y(x^2-y^2)}{(x^2+y^2)^2}$$

⑥ we use the definition:

$$f_{xy}(0,0) = \frac{\partial f_y}{\partial x}(0,0) = \lim_{h \rightarrow 0} \frac{f_y(h,0) - f_y(0,0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h \cdot \arctan 1}{h} = \arctan 1 = \frac{\pi}{4}$$

$$f_{yx}(0,0) = \frac{\partial f_x}{\partial y}(0,0) = \lim_{h \rightarrow 0} \frac{f_x(0,h) - f_x(0,0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h \arctan(-1)}{h} = \arctan(-1) = -\frac{\pi}{4}$$

© False since $f_{xy}(0,0) \neq f_{yx}(0,0)$



definition: Let $\vec{0} \neq \vec{u} \in \mathbb{R}^n$. the directional derivative of a fun. $\vec{F}$ at $\vec{x}_0$ in the direction $\vec{u}$ is defined as

$$D_{\vec{u}}(\vec{F})(\vec{x}_0) = \lim_{h \rightarrow 0} \frac{\vec{F}(\vec{x}_0 + h\vec{u}) - \vec{F}(\vec{x}_0)}{h \cdot \|\vec{u}\|}$$

Remark 1 We have $D_{\vec{e}_j}(\vec{F}) = \frac{\partial \vec{F}}{\partial x_j}$. thus the directional derivative generalizes partial derivatives.

Remark 2: note that for $\vec{v} = \frac{\vec{u}}{\|\vec{u}\|}$ we have

(i) $\|\vec{v}\| = 1$

(ii) $D_{\vec{v}}(\vec{F})(\vec{x}_0) = D_{\vec{u}}(\vec{F})(\vec{x}_0)$ since

$$D_{\vec{u}}(\vec{F})(\vec{x}_0) = \lim_{h \rightarrow 0} \frac{\vec{F}(\vec{x}_0 + h\vec{u}) - \vec{F}(\vec{x}_0)}{h \cdot \|\vec{u}\|} = \lim_{t \rightarrow 0} \frac{\vec{F}(\vec{x}_0 + t \cdot \frac{\vec{u}}{\|\vec{u}\|}) - \vec{F}(\vec{x}_0)}{t}$$

$$= \lim_{t \rightarrow 0} \frac{\vec{F}(\vec{x}_0 + t \vec{v}) - \vec{F}(\vec{x}_0)}{t} = D_{\vec{v}}(\vec{F})(\vec{x}_0)$$

thus the directional derivative only depends on the direction of $\vec{u}$, but not on its norm.

Exercise 7: Consider the fun.

$$F(x,y) = \begin{cases} \frac{xy}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$$

compute $D_{\vec{u}}(\vec{F})(0,0)$ for $\vec{u} = (1,1)$.

Solution: we have

$$D_{\vec{u}}(F)\begin{pmatrix} 0 \\ 0 \end{pmatrix} = \lim_{h \rightarrow 0} \frac{F\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix} + h \begin{pmatrix} 1 \\ 1 \end{pmatrix}\right) - F\begin{pmatrix} 0 \\ 0 \end{pmatrix}}{h \cdot \|\vec{u}\|} =$$

$$= \lim_{h \rightarrow 0} \frac{F\begin{pmatrix} h \\ h \end{pmatrix}}{h \cdot \sqrt{2}} = \lim_{h \rightarrow 0} \frac{\left(\frac{h^2}{2h^2}\right)}{h \cdot \sqrt{2}} = \lim_{h \rightarrow 0} \frac{1}{h \cdot 2\sqrt{2}}$$

= doesn't exist.



definition: let $D \subseteq \mathbb{R}^n$ and let $\vec{F}: D \rightarrow \mathbb{R}^m$. Let

$\vec{x}_0 \in D$. The total derivative of $\vec{F}$ is a matrix of size $m \times n$, denoted by $D_{\vec{F}}(\vec{x}_0) = D(\vec{F})(\vec{x}_0)$

and defined by

$$[D_{\vec{F}}(\vec{x}_0)]_{ij} = \frac{\partial F_i}{\partial x_j}(\vec{x}_0), \quad \forall \begin{matrix} 1 \leq i \leq m \\ 1 \leq j \leq n \end{matrix}$$

$$\text{where } \vec{F} = \begin{pmatrix} F_1 \\ \vdots \\ F_m \end{pmatrix}$$

Exercise 8: Compute the derivative of each of the following functions:

(a) $f(x,y) = x^2 + y^3$

solution:

$$D_f(x,y) = (2x, 3y^2)$$



(b) $f(x,y) = \begin{pmatrix} x^2 \\ y^2 \end{pmatrix}$

solution:

$$D_f(x,y) = \begin{pmatrix} 2x & 0 \\ 0 & 2y \end{pmatrix}$$

(c) $f(x,y,z) = \begin{pmatrix} 1+x+z^2 \\ 1-y \\ xz \end{pmatrix}$

solution:

$$D_f(x,y,z) = \begin{pmatrix} 1 & 0 & 2z \\ 0 & -1 & 0 \\ z & 0 & x \end{pmatrix}$$



$$\textcircled{d} \quad \vec{F}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} xy \\ 2x \\ 3y \end{pmatrix} \quad \forall x, y \in \mathbb{R}.$$

solution:

$$D_{\vec{F}}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y & x \\ 2 & 0 \\ 0 & 3 \end{pmatrix}$$

$$\textcircled{e} \quad \vec{G}\begin{pmatrix} u \\ v \\ w \end{pmatrix} = \begin{pmatrix} u+3v \\ 2v-uvw \end{pmatrix}, \quad \forall u, v, w \in \mathbb{R}.$$

solution:

$$D_{\vec{G}}\begin{pmatrix} u \\ v \\ w \end{pmatrix} = \begin{pmatrix} 1 & 3 & 0 \\ 0 & 2-w & -v \end{pmatrix}$$



Exercise 9: Let $x_0 \in \mathbb{R}$. Let f be a fun. defined on a neigh. of x_0 . prove: f is diff. at x_0 iff $\exists a \in \mathbb{R}$ s.t.

$$\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0) - a(x - x_0)}{|x - x_0|} = 0$$

moreover, prove that in this case $a = f'(x_0)$.

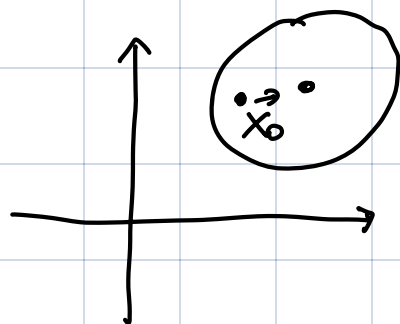
solution: HW11 Q2 in calculus 1.



definition: Let $\vec{x}_0 \in \mathbb{R}^n$ and let $r > 0$. An open ball of radius r centered at $\vec{x}_0$ is defined as the set

$$B_r(\vec{x}_0) = \{ \vec{x} \in \mathbb{R}^n : \|\vec{x} - \vec{x}_0\| < r \}$$

A neighborhood of $\vec{x}_0$ is an open ball that contains $\vec{x}_0$ (but not necessarily at its center)



definition: Let $D \subseteq \mathbb{R}^n$ and let $\vec{F}: D \rightarrow \mathbb{R}^m$.

Let $\vec{x}_0 \in \mathbb{R}^n$ and suppose that $\vec{F}$ is defined at a neigh. of $\vec{x}_0$. We say that $\vec{F}$ is

diff. at $\vec{x}_0$ if $\exists A \in M_{m \times n}$ s.t

$$\lim_{\vec{x} \rightarrow \vec{x}_0} \frac{\vec{F}(\vec{x}) - \vec{F}(\vec{x}_0) - A \cdot (\vec{x} - \vec{x}_0)}{\|\vec{x} - \vec{x}_0\|} = \vec{0}$$

Theorem 2: If F is diff. at $\vec{x}_0$ then

$$D_{\vec{F}}(\vec{x}_0) \text{ exists and } A = D_{\vec{F}}(\vec{x}_0)$$

proof: No proof. 😊

Theorem 3: If $\frac{\partial \vec{F}}{\partial x_j}$ are cont. at $\vec{x}_0$
 $\forall 1 \leq j \leq n$, then F is diff. at $\vec{x}_0$.

proof: No proof. 😊

Exercise 11: consider the fun. defined by

$$F\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{cases} \frac{xy}{x^2+y^2} & \begin{pmatrix} x \\ y \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\ 0 & \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \end{cases}, \forall \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2$$

(a) compute $D_F\left(\begin{pmatrix} x \\ y \end{pmatrix}\right)$ at every point $\begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2$

(b) Determine whether F is diff at $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$

solution: we already saw

$$\frac{\partial F}{\partial x} \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \frac{\partial F}{\partial y} \begin{pmatrix} 0 \\ 0 \end{pmatrix} = 0. \quad \forall \begin{pmatrix} x \\ y \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix} \text{ we can}$$

use AOD:

$$\text{thus} \quad \frac{\partial F}{\partial x} \begin{pmatrix} x \\ y \end{pmatrix} = \frac{y \cdot (x^2 + y^2) - 2x^2 y}{(x^2 + y^2)^2}$$

Hence

$$\frac{\partial F}{\partial x} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{cases} \frac{y \cdot (x^2 + y^2) - 2x^2 y}{(x^2 + y^2)^2} & \begin{pmatrix} x \\ y \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\ 0 & \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \end{cases}$$

and similarly, since $F(x, y)$ is symmetric,

$$\frac{\partial F}{\partial y} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{cases} \frac{x \cdot (x^2 + y^2) - 2y^2 x}{(x^2 + y^2)^2} & \begin{pmatrix} x \\ y \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\ 0 & \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \end{cases}$$

thus

$$D_F \begin{pmatrix} x \\ y \end{pmatrix} = \left(\frac{y \cdot (x^2 + y^2) - 2x^2 y}{(x^2 + y^2)^2}, \frac{x \cdot (x^2 + y^2) - 2y^2 x}{(x^2 + y^2)^2} \right)$$

$$\forall \begin{pmatrix} x \\ y \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix} \text{ and}$$

$$D_F \begin{pmatrix} 0 \\ 0 \end{pmatrix} = (0, 0)$$

(b) By theorem 2, $A = D_F \begin{pmatrix} 0 \\ 0 \end{pmatrix} = (0, 0)$.

Hence we compute by def:

$$\lim_{\begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 0 \end{pmatrix}} \frac{F \begin{pmatrix} x \\ y \end{pmatrix} - F \begin{pmatrix} 0 \\ 0 \end{pmatrix} - (0, 0) \cdot \left(\begin{pmatrix} x \\ y \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right)}{\left\| \begin{pmatrix} x \\ y \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\|}$$

$$= \lim_{\begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 0 \end{pmatrix}} \frac{\frac{xy}{x^2+y^2} - 0}{\sqrt{x^2+y^2}} = \lim_{\begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 0 \end{pmatrix}} \frac{\frac{xy}{x^2+y^2}}{(x^2+y^2)^{3/2}}$$

$$= \lim_{r \rightarrow 0^+} \frac{r^2 \cos \theta \sin \theta}{(r^2)^{3/2}} = \lim_{r \rightarrow 0^+} \frac{\cos \theta \sin \theta}{r}$$

= does not exist!

definition: Let $D \subseteq \mathbb{R}^n$ and let $F: D \rightarrow \mathbb{R}$.

Let $\vec{x}_0 \in \mathbb{R}^n$ and suppose that F is defined at a neigh. of $\vec{x}_0$. The gradient of F is defined as

$$\begin{aligned}\nabla F(\vec{x}_0) &= [D_F(\vec{x}_0)]^t \\ &= \left(\frac{\partial F}{\partial x_1}, \dots, \frac{\partial F}{\partial x_n} \right)^t = \begin{pmatrix} \frac{\partial F}{\partial x_1} \\ \vdots \\ \frac{\partial F}{\partial x_n} \end{pmatrix}\end{aligned}$$

Example:

$$F \begin{pmatrix} x \\ y \\ z \end{pmatrix} = xy + z^2x + y.$$

$$\nabla F = \begin{pmatrix} y + z^2 \\ x + 1 \\ 2xz \end{pmatrix}$$

$$D_F = (y + z^2, x + 1, 2xz)$$

Theorem 4: Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$, and let $\vec{u} \in \mathbb{R}^2$ be a unit vector. If f_x, f_y are cont. at $\vec{x}_0$, then

$$D_{\vec{u}}(f)(\vec{x}_0) = \langle \nabla f(\vec{x}_0), \vec{u} \rangle \\ = f_x \cdot a + f_y \cdot b \quad \text{where } \vec{u} = \begin{pmatrix} a \\ b \end{pmatrix}.$$

Example: Let $f(x, y) = x \cdot y^3, \forall (x, y) \in \mathbb{R}^2$.

and let $u = (2, 3)$.

compute $D_{\vec{u}}(f)(x, y)$ for every $(x, y) \in \mathbb{R}^2$

Solution: we have

$$\nabla f \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y^3 \\ 2xy \end{pmatrix}$$

$$\frac{\vec{u}}{\|\vec{u}\|} = \frac{1}{\sqrt{13}} \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

$$D_{\vec{u}}(f)(x, y) = \left\langle \begin{pmatrix} y^3 \\ 2xy \end{pmatrix}, \frac{1}{\sqrt{13}} \begin{pmatrix} 2 \\ 3 \end{pmatrix} \right\rangle$$

$$= \frac{1}{\sqrt{13}} \cdot (2y^3 + 6xy)$$

